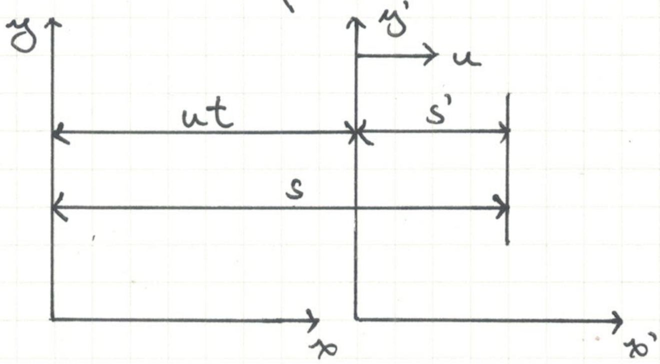
Topic A1 : Inertial frames

Inertial frames of reference : Frame of reference in which $F_{net} = 0$.

Principle of Relativity : Laws of physics hold true in any inertial frames of reference.

Frame of reference : System of coordinates for space and time.

Galilean transformation :



$$y = y'$$
$$z = z'$$
$$t = t'$$
$$s = s' + ut \Rightarrow v = v' + u$$

Centre of mass

Position of centre of mass

$$X_{cm} = \frac{m_1x_1 + m_2x_2 + m_3x_3 + \cdots}{m_1 + m_2 + m_3 + \cdots} = \frac{\sum_i m_i x_i}{\sum_i m_i}$$

$$\Rightarrow M\vec{r}_{cm} = \sum_i m_i \vec{r}_i$$

$M = \sum_i m_i$

$\vec{r}$ is the respective positional vectors

Motion of centre of mass

Differentiating X_{cm} w.r.t. t,

$$M\vec{v}_{cm} = \sum_i m_i \vec{v}_i$$
$$= \vec{P}$$

Differentiating $\vec{v}_{cm}$ w.r.t. t,

$$\sum \vec{F} = \sum \vec{F}_{ext} + \sum \vec{F}_{int} = M\vec{a}_{cm} = \sum_i m_i \vec{a}_i$$

Since $\sum \vec{F}_{int} = 0$ due to N3L,

$$\sum \vec{F}_{ext} = M\vec{a}_{cm}$$

* For elastic collisions, velocities in CM frame simply change direction.

Topic A2 : Rotational Motion

Kinematics & Dynamics

Translational motion	Rotational motion
Displacement, s	Angular displacement, θ
Velocity, $v = \frac{ds}{dt}$	Angular velocity, $\omega = \frac{d\theta}{dt}$
Acceleration, $a = \frac{dv}{dt}$	Angular acceleration, $\alpha = \frac{d\omega}{dt}$
Mass, $m = \sum_{n=1}^{N} m_n$	Moment of inertia, $I = \sum_{n=1}^{N} m_n r^2 = \int r^2 \, dm$
Linear momentum, $p = mv$	Angular momentum, $L = I\omega$
Force, $F = \frac{dp}{dt} = ma$	Torque, $\tau = \frac{dL}{dt} = I\alpha$
$KE = \frac{1}{2} m v^2$	$KE = \frac{1}{2} I \omega^2$
$Work = \int_{x_i}^{x_f} F \, dx$	$Work = \int_{\theta_i}^{\theta_f} \tau \, d\theta$
$Power = Fv$	$Power = \tau \omega$

Equations:

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$$

$$\omega = \omega_0 + \alpha t$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

$$\theta = \frac{\omega_0 + \omega}{2}(t)$$

Parallel-axis theorem : $I = I_{cm} + Mh^2$

h is the perpendicular distance between the axises.

If μ_s is sufficiently large, point of contact does not move, energy is conserved.

For an object rolling without slipping,

$$\text{Total KE} = \text{Rotational KE about centre of mass} + \text{Translational KE}$$

$$= \frac{1}{2} I_{cm}\omega^2 + \frac{1}{2} M (R\omega)^2$$

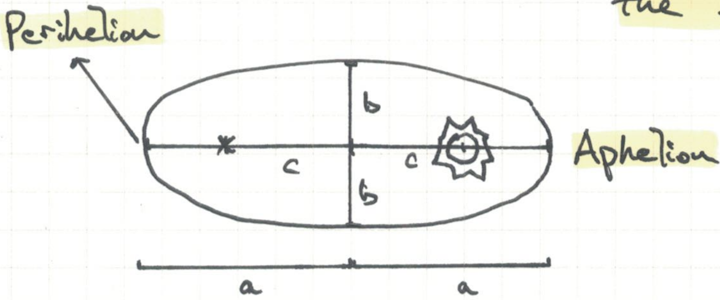
Friction is $\propto$ Normal contact force, <u>independent</u> of area of contact.

$f = \mu N$

Topic A3 : Planetary and Satellite Motion

Kepler's laws of planetary motion

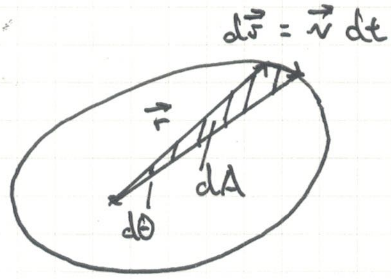
Kepler's first law : Planets move in elliptical orbits with the sun at one focus of the ellipse.



$$c = ea$$

$$c = \sqrt{a^2 - b^2}$$

Kepler's second law : An imaginary line from the sun to a given planet sweeps out equal areas in equal intervals of time.



* Gravitational force always acts towards the focus. Hence, net torque on the planet is zero and angular momentum remains constant.

$$dA = \frac{1}{2}\left|\vec{r} \times d\vec{s}\right| = \frac{1}{2}\left|\vec{r} \times \vec{r}\,d\theta\right|$$

Sector velocity $\dfrac{dA}{dt} = \dfrac{1}{2}\left|\vec{r} \times \vec{r}\,\dfrac{d\theta}{dt}\right|$

$$= \frac{1}{2} r^2 \omega$$

$$= \frac{L}{2m}$$, where m is the mass of planet

Sector velocity is constant

$\Rightarrow$ Angular momentum L is constant !

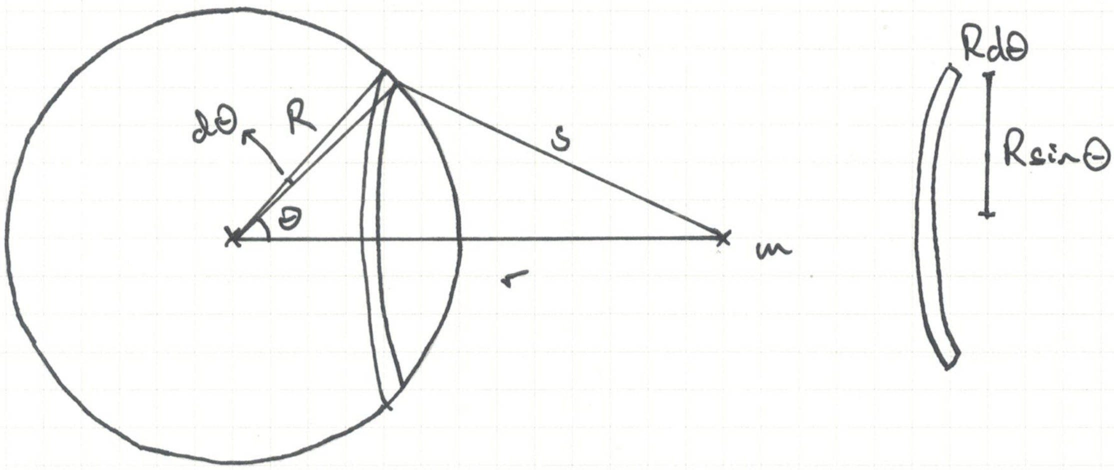
Kepler's third law : Ratio of the square of a planet's period to the cube of the semi-major axis of its orbit around the sun is a constant and this constant is the same for all planets.

$$T^2 = \frac{4\pi^2}{GM} a^3$$

* If T is in years,
  M is in solar mass,
  and a is in AU, $T^2 = a^3$

Gravitational potential energy

G.P.E. of spherical shell :



Let mass per unit area $= \sigma$

$M = \sigma 4\pi R^2$

$\Rightarrow \sigma = \dfrac{M}{4\pi R^2}$

Mass of strip $dM = \sigma\,(\text{width}) \times (\text{length of strip})$

$= \sigma R\,d\theta\ \pi\ 2R\sin\theta$

$= \sigma 2\pi R^2\,d\theta\sin\theta$

$dU = -\dfrac{Gm\,dM}{s}$

$= -\dfrac{Gm\sigma 2\pi R^2\,d\theta\sin\theta}{s}$ (sub dM)

$= -\dfrac{GmM\,2\pi R^2\,d\theta\sin\theta}{4\pi R^2\,s}$ (sub σ)

$= -\dfrac{GmM}{2s}\,d\theta\sin\theta$

Using Law of Cosines, $s^2 = R^2 + r^2 - 2Rr\cos\theta$

$\Rightarrow 2s\,ds = 2Rr\sin\theta\,d\theta$

$d\theta\sin\theta = \dfrac{s\,ds}{Rr}$

$dU = -\dfrac{GmM}{2Rr}\,ds$

U outside shell : $U = -\dfrac{GmM}{2Rr}\displaystyle\int_{r-R}^{r+R} ds = -\dfrac{GmM}{2Rr}\left[r+R-(r-R)\right]$

$= -\,GmM$

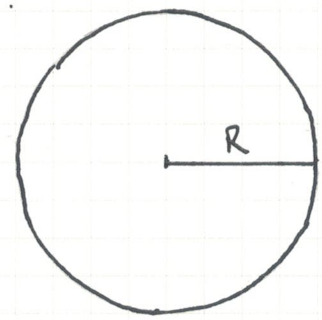
U inside shell : $U = -\frac{GmM}{2Rr}\int_{R-r}^{R+r} ds$

$$= -\frac{GmM}{2Rr}\left[R+r-(R-r)\right]$$

$$= -\frac{GmM}{R} \quad (\text{Constant})$$

$\Rightarrow$ Inside a spherical shell, mass have constant GPE and does not experience F_G.

<u>G.P.E. within a solid, uniform sphere</u> :



Let mass per unit volume be ρ,

$$M = \rho \frac{4}{3}\pi R^3$$

$$\rho = \frac{M}{\frac{4}{3}\pi R^3}$$

$$dU = -\frac{GmdM}{r}$$

Considering a hollow spherical shell of dM,

$$dM = \rho\, dV = \rho\, 4\pi r^2 dr$$

$$dU = -\frac{Gm\rho 4\pi r^2 dr}{r}$$

Sub ρ,

$$= -\frac{GmM\,4\pi r^2 dr}{\frac{4}{3}\pi R^3\, r}$$

$$= -\frac{3GmM}{R^3} r\, dr$$

$$U \text{ at center} = -\frac{3GmM}{R^3}\int_0^R r\, dr$$

$$= -\frac{3GmM}{2R^3}\left[r^2\right]_0^R$$

$$= -\frac{3GmM}{2R}$$

Elliptical orbits and orbit transfer

For an object of mass m orbiting mass M,

$$KE_{Total} = KE_{Radial} + KE_{Tangential}$$

$$= \frac{1}{2} m v_r^2 + \frac{1}{2} m v_\perp^2$$

Sub $L = m r v_\perp$,

$$= \frac{1}{2} m v_r^2 + \frac{L^2}{2mr^2}$$

$$E_{Total} = KE + GPE$$

$$= \frac{1}{2} m v_r^2 + \boxed{\frac{L^2}{2mr^2} - \frac{GMm}{r}}$$

Effective radial potential, $U_{eff} = \dfrac{L^2}{2mr^2} - \dfrac{GMm}{r}$

* U_{eff} is only dependent on r and m.

Energy in elliptical orbits at turning point ($v_r = 0$):

$$E_{Total} = U_{eff}$$

Equating $\dfrac{L^2}{2m}$ at nearest and furthest separation,

$$E_{Total} r_{min}^2 + GMm r_{min} = E_{Total} r_{max}^2 + GMm r_{max}$$

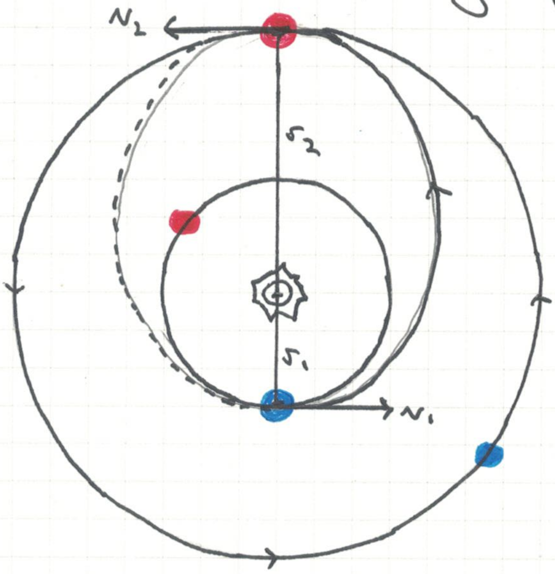
$$\Rightarrow E_{Total} = -\frac{GMm}{r_{max} + r_{min}}$$

$$= -\frac{GMm}{2a}$$

Total energy of object in elliptical orbit is dependent on the major axis, M and m.

Hohmann transfer

↳ Most economical way of interplanetary transfer.



● : Earth and Mars at departure

● : Earth and Mars at arrival

⇝ Two engine burns are required.

Burn 1 : Circular Earth orbit v_1 $\xrightarrow{\Delta v_1}$ Elliptical transfer orbit v_{e_1}

Burn 2 : Elliptical transfer orbit v_{e_2} $\xrightarrow{\Delta v_2}$ Circular Mars orbit v_2

Let mass of sun be M,

Earth orbit :

$$E_{Total} = -\frac{GMm}{2r_1} = \frac{1}{2}mv_1^2 - \frac{GMm}{r_1}$$

$$\frac{1}{2}mv_1^2 = \frac{GMm}{2r_1}$$

$$v_1 = \sqrt{\frac{GM}{r_1}}$$

Elliptical orbit (~~ap~~ perihelion) :

$$E_{total} = -\frac{GMm}{r_1+r_2} = \frac{1}{2}mv_{t_1}^2 - \frac{GMm}{r_1}$$

$$v_{t_1} = \sqrt{\frac{2GM}{r_1+r_2}\frac{r_2}{r_1}}$$

$$\Delta v_1 = v_{t_1} - v_1$$

Elliptical orbit (apehelion) :

$$v_{t_2} = \sqrt{\frac{2GM}{r_1+r_2}\frac{r_1}{r_2}}$$

Mars orbit :

$$v_2 = \sqrt{\frac{GM}{r_2}}$$

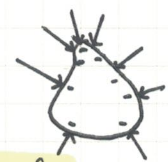
$$\Delta v_2 = v_2 - v_{t_2}$$

Topic B1: Electric and Magnetic fields

Electric field

- $E = \frac{F}{q_0} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$ (Point charge)
- $\vec{E}_{net} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \cdots$ (Principle of superposition of electric fields)
- Electric fields in a conductor

1. The surface of any charged conductor in electrostatic equilibrium is equipotential.
2. Electric field within charged conductor is zero.
   → Consider a gaussian surface within the conductor. Since the charge it encloses is zero, the electric field must be zero.
3. All electric charge accumulates on the surface of the conductor.
4. Surface charge density is greatest at locations where the radius of curvature of the surface is smallest.



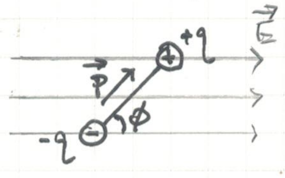
- Gauss's Law

Electric flux $\Phi = \oint \vec{E} \cdot d\vec{A} = EA = \frac{q_{enclosed}}{\varepsilon_0}$

Since E is constant!

* If charge density changes with radius, integrate with infinitesimal volume to find q_{encl}.

- Electric dipoles : A pair of point charges with equal magnitude and opposite sign separated by distance d.



→ $\tau = F d_{\perp} = (qE)(d \sin\phi)$
⇒ $\vec{\tau} = \vec{p} \times \vec{E}$

→ Electric dipole moment $\vec{p}$: Product of charge q and distance d
$\vec{p} = qd$ Units: C·m Direction: From ⊖ to ⊕

→ Potential energy

Work done $= \int -pE\sin\phi \, d\phi$ (Negative since torque is in direction of decreasing ϕ)
$= pE\cos\phi$

Since W = Decrease in U

$U = -pE\cos\phi = -\vec{p}\cdot\vec{E}$

$\phi = 0°, 180° \Rightarrow \tau = 0, U = -\vec{p}\cdot\vec{E}$ / $\vec{p}\cdot\vec{E}$

$\phi = 90° \Rightarrow \tau = pE = qEd, U = 0$

Magnetic field

↳ Ampere's Law

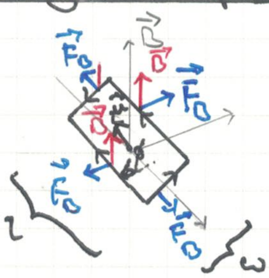
Line integral $\oint \vec{B} \cdot d\vec{s} = \frac{\mu_0 I}{2\pi r}(2\pi r)$

$= \mu_0 I_{enclosed}$

* Total integral of $\vec{B}$ around the loop is zero. However, $\vec{B}$ may vary along the loop.

* If $\oint \vec{B} \cdot d\vec{s} = 0$, $I_{enclosed} = 0$.

↳ Magnetic dipole : A current carrying closed loop.



↳ $\tau = F d_{\perp} = 2(BIl)(\frac{w}{2} \sin\phi)$

$= BIA \sin\phi$

$\Rightarrow \vec{\tau} = \vec{\mu} \times \vec{B}$

→ Magnetic dipole moment $\vec{\mu}$: Product of current and area of the loop.

$\vec{\mu} = I\vec{A}$ Units: $A \cdot m^2$

Direction: Perpendicular to the area, following right hand grip rule.

* $\vec{\mu}$ is independent of shape of loop shape if area is the same.

→ Potential energy

$U = -\mu B \cos\phi = -\vec{\mu} \cdot \vec{B}$

$\phi = 0°, 180° \Rightarrow \tau = 0,\ U = -\cancel{pE}\,\mu B,\ \cancel{pE}\,\mu B$

$\phi = 90° \Rightarrow \tau = \mu B,\ U = 0$

* Net force on a magnetic dipole in a uniform magnetic field is zero, but net torque is not in general equal to zero.

↳ Gauss's Law

$\oint \vec{B} \cdot d\vec{A} = 0$

→ Since magnetic fields are continuous and form closed loops, number of field lines entering and exiting the surface is equal.
Net magnetic flux is zero.

→ Hence, there exist no magnetic monopoles to act as sources of magnetic field.

Topic B2: Capacitors and Inductors

Capacitors $\dashv\vdash$

→ Capacitance : Ratio of charge accumulated on each conductor to the voltage between the conductors.

$$C = \frac{Q}{V}$$ Units: Farad F

* The larger the capacitance, the larger the amount of charge able to be stored for a given potential difference.
Hence, capacitance is a measure of the capacitor's ability to store energy.

→ Parallel-plates capacitors

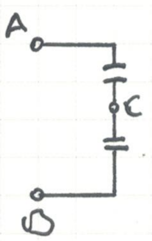
$$E = \frac{Q}{\varepsilon_0 A}$$

$$V = Ed = \frac{Q}{\varepsilon_0 A} d$$

$$\Rightarrow C = \frac{Q}{V} = \frac{\varepsilon_0 A}{d}$$

→ Combination of capacitors

Series



$$V_{AB} = V_{AC} + V_{CB}$$
$$= V_1 + V_2$$
$$= \frac{Q}{C_1} + \frac{Q}{C_2}$$

$$\Rightarrow C_{eff} = \frac{Q}{V_{AB}} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2}}$$

$$\Rightarrow \frac{1}{C_{eff}} = \sum_i \frac{1}{C_i}$$

Parallel

$$Q_1 = C_1 V \quad , \quad Q_2 = C_2 V$$

$$C_{eff} = \frac{Q}{V}$$
$$= \frac{Q_1 + Q_2}{V}$$

$$\Rightarrow C_{eff} = \sum_i C_i$$

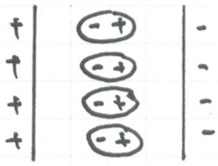
→ Energy

$$V = \frac{dW}{dq} \Rightarrow dW = V\,dq = \frac{Q}{C}\,dq$$

$$U = \int_0^Q \frac{q}{C}\,dq = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}QV = \frac{1}{2}V^2 C$$

* Work done is area under V-q graph.
* Energy is stored in the electric field.
Energy is the work done to separate charges onto the two plates.

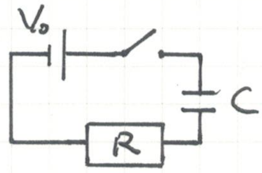
↳ Dielectric materials



For the same amount of charge stored, the voltage across the plates is lower.

$C = \frac{Q}{V}$ $\Rightarrow$ C increases

↳ RC circuits



Charging :

Voltage across must be equal e.m.f.

$$V_0 = V_R + V_C = IR + \frac{Q}{C}$$

$$V_0 = \frac{dQ}{dt} R + \frac{Q}{C}$$

$$\Rightarrow \frac{dQ}{dt} = \frac{V_0}{R} - \frac{1}{RC} Q$$

Differentiating both sides w.r.t. t,

$$\frac{dI}{dt} = -\frac{1}{RC} I$$

$$\int_0^{I_0} \frac{1}{I} dI = -\int \frac{1}{RC} dt$$

$$\Rightarrow \boxed{\begin{aligned} I &= I_0 e^{-\frac{1}{RC}t} \\ &= \frac{V_0}{R} e^{-\frac{1}{RC}t} \end{aligned}}$$

(Since at $t=0$, $Q=0$, $V_0 = V_R = IR$)

$$Q = \int I \, dt$$

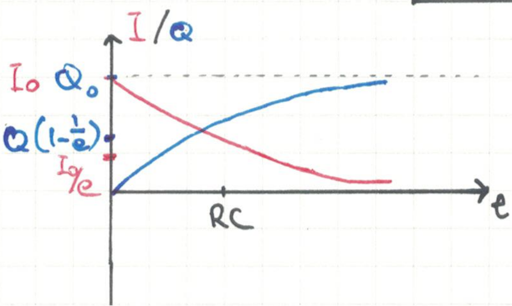
$$= -V_0 C e^{-\frac{1}{RC}t} + c$$

Since at $t=0$, $Q=0$, $c = V_0 C$

$$\Rightarrow \boxed{Q = V_0 C \left(1 - e^{-\frac{1}{RC}t}\right)}$$

Sub $V = \frac{Q}{C}$,

$$\Rightarrow \boxed{V = V_0 \left(1 - e^{-\frac{1}{RC}t}\right)}$$



Discharging:

$$V_R + V_c = 0$$

$$IR + \frac{Q}{C} = 0$$

$$\frac{dQ}{dt}R + \frac{Q}{C} = 0$$

$$\frac{dQ}{dt} = \cancel{0} - \frac{1}{RC}Q$$

$$\int_{Q_0}^{Q} \frac{1}{Q}\,dQ = -\int \frac{1}{RC}\,dt$$

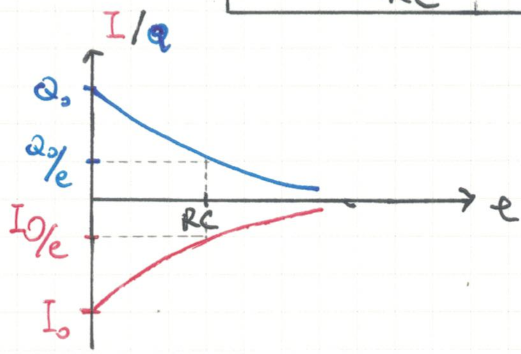
$$\Rightarrow \boxed{Q = Q_0 e^{-\frac{1}{RC}t}}$$ (since at $t=0$, $Q=Q_0$)

Sub $V = \frac{Q}{C}$,

$$\Rightarrow \boxed{V = V_0 e^{-\frac{1}{RC}t}}$$

Sub $I = \frac{dQ}{dt}$,

$$\Rightarrow \boxed{I = \frac{Q_0}{RC} e^{-\frac{1}{RC}t} = I_0 e^{-\frac{1}{RC}t}}$$



⇒ Since (RC) determines how quickly a capacitor is charged and discharged, (RC) is known as the time constant.

Inductors

→ **Inductance** : Tendency of a conductor to oppose a change in th electrical current flowing through it.

Self-inductance

$$e.m.f = -L\frac{dI}{dt}$$ Units: Henry H

→ **Ideal solenoid inductors**

$$B = \mu_0 n I = \frac{\mu N I}{l}$$

$$e.m.f = -L\frac{dI}{dt} = -N\frac{d\phi}{dt}$$

$$= -\frac{d(NBA)}{dt}$$

$$-L\frac{dI}{dt} = -\frac{\mu_0 N^2 A}{l}\frac{dI}{dt}$$

$$\Rightarrow L = \frac{\mu_0 N^2 A}{l}$$

→ Combination of inductors

Series

By conservation of charge,

$$e.m.f. = \sum_i (V_L)_i$$

$$= -\frac{dI}{dt}\left(\sum_i L_i\right)$$

$$\Rightarrow L_{eff.} = \sum_i L_i$$

Parallel

By conservation of current,

$$I_{Total} = \sum_i I_i$$

Differentiating w.r.t. time,

$$\frac{dI_{Total}}{dt} = \sum_i \frac{dI_i}{dt}$$

$$\frac{e.m.f}{L_{eff}} = \sum_i \frac{(e.m.f)_i}{L_i}$$

$$\Rightarrow \frac{1}{L_{eff}} = \sum_i \frac{1}{L_i}$$

↳ Energy

$$P = IV = IL\frac{dI}{dt}$$

$$E = \int_0^T P\,dt$$

$$= \int_0^{I_0} I L\,dI$$

$$U = \frac{1}{2} L I_0^2$$

* Work done is area under L-I graph.

* Energy is stored in magnetic field.

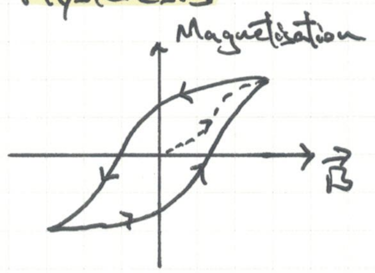
↳ Ferromagnetism and hysteresis

→ Ferromagnetic materials have atoms that align their spin to the externally applied magnetic field.

For the same applied current, magnetic field stronger, and back e.m.f is stronger.

Inductance increases.

→ Hysteresis



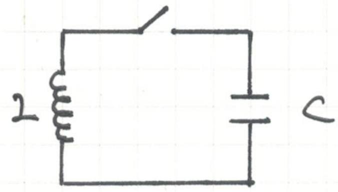
→  : As the external $\vec{B}$ field increases, the $\vec{B}$ field of material increases with diminishing returns, as less and less atoms are available for alignment.

* Hysteresis loop shows the "history dependent" nature of magnetisation in ferromagnetic materials.

→  : When driving $\vec{B}$ drops to zero, ferromagnetic material retains much of the magnetisation.

The driving magnetic field must be reversed and increased to a large value to drive the magnetisation to zero, and saturation in the opposite direction.

↳ LC oscillator circuit



$$V_C - V_L = 0$$

$$\frac{Q}{C} - \left(-L\frac{dI}{dt}\right) = 0$$

$$\Rightarrow \frac{d^2Q}{dt^2} + \frac{1}{LC}Q = 0$$

$$\frac{d^2Q}{dt^2} = -\frac{1}{LC}Q$$

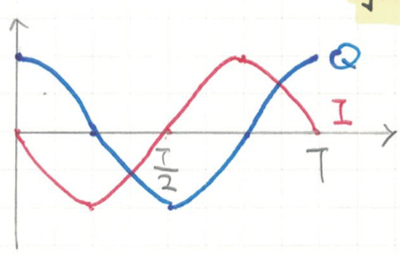
* Similar to simple harmonic motion.

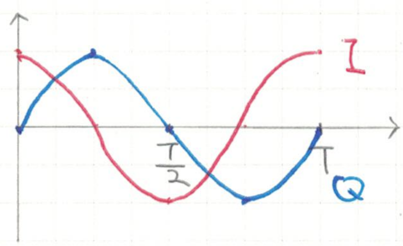
$$a = \frac{d^2x}{dt^2} = -kx$$

At $t = 0$,

Max Q	Max I
$Q = Q_0 \cos(\omega t + \phi)$	$Q = Q_0 \sin(\omega t + \phi)$
$I = -I_0 \sin(\omega t + \phi)$	$I = I_0 \cos(\omega t + \phi)$

, where $\omega = \frac{1}{\sqrt{LC}}$





* Energy graph is SQUARED!

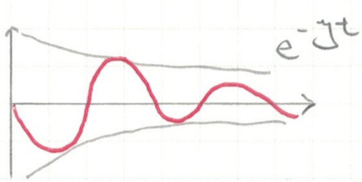
↳ RLC damped oscillator circuit

$$V_L - V_C - V_R = 0$$

$$-L\frac{dI}{dt} = \frac{Q}{C} + IR$$

$$\Rightarrow \frac{d^2Q}{dt^2} + \frac{R}{L}\frac{dQ}{dt} + \frac{1}{LC}Q = 0$$

* Frequency of damped oscillation depends on R, L and C.
As R increases, damping increases. Period ↑, frequency ↓.



* For light damping, ω must be real. ($\omega > 0$)
For critical damping, ω must be 0.
For heavy damping, ω must be imaginary.